



# Linear Relationships

## Topic Test 1 Solution

Total Marks: 30

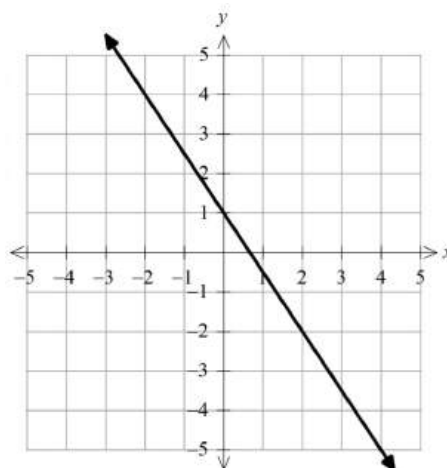
Time: 50 minutes

- 1 What is the gradient and y-intercept of the line  $y = 7 - x$ ? 1

$m = -1$  y-intercept is  $(0, 7)$

- 2 Find the equation of the line below. 2

The linear relationship's graph shows a negative gradient of  $\frac{3}{2}$  and y-intercept of 1. The equation is  $y = -\frac{3x}{2} + 1$



- 3 The volume ( $V$  litres) of water in a tank as it is being drained over time ( $t$ ) minutes can be modelled by the equation  $V = 360 - 12t$ .

- a. At what rate is the tank draining? Gradient is 12 therefore 12 L/min 1
- b. What is the initial volume of the tank? 1

Y-intercept is 360 therefore initial volume is 360 litres.

- c. After how many minutes will the tank have half the initial volume of water? 2

$$360 - 12t = 180$$

$$12t = 180 \quad t = 15$$

- 4 A hairdresser charges a \$35 fee for a haircut as well as \$0.75 per additional minute used to colour a client's hair. If a client's colour takes  $t$  hours, which formula could be used to calculate the total charge (\$C)? 2

$$\text{Cost for cut and colour is } \$35 + 0.75 \times 60t$$

$$C = 35 + 45t$$

- 5 A preschool centre pays \$210 to hire a bus for an excursion. Let  $C$  be the cost per person and  $N$  be the number of people going on the excursion. Write an equation connecting  $C$  and  $N$ , if the cost per person is \$2.10. Hence, determine the number of people who travelled on the bus if the centre paid \$315. 2

$$315 = 210 + 2.10N$$

$$315 - 210 = 105 = 2.10N$$

$$2.10 \quad 50 = N$$

- 6 The table below shows the speed  $v$  in  $km/s$  of a plane at time  $t$  seconds.

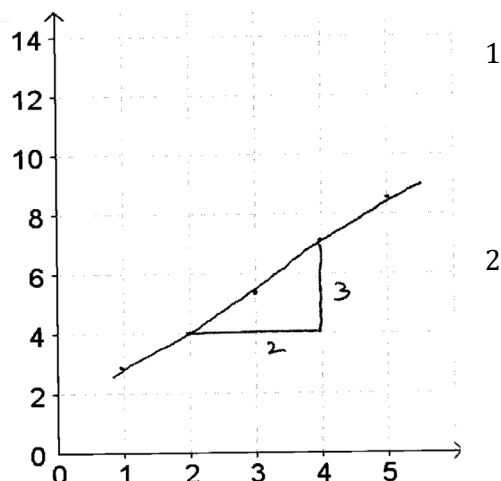
| Time ( $t$ )  | 1   | 2 | 3   | 4 | 5   |
|---------------|-----|---|-----|---|-----|
| Speed ( $v$ ) | 2.5 | 4 | 5.5 | 7 | 8.5 |

- a. Draw a number plane with  $t$ (time) as the horizontal axis and  $v$  (speed) as the vertical axis. Plot the points and join them to make a straight line.
- b. Determine the equation of the line in the form  $y = m + c$ .

$$y = m + c.$$

$$...m = 1.5...$$

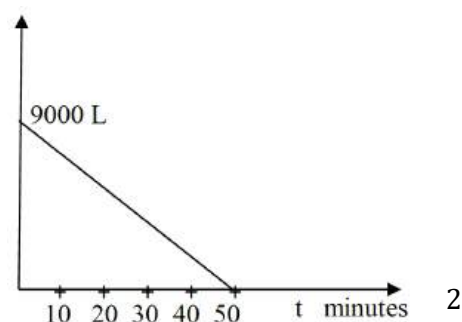
$$...y = 1.5x + 1.$$



- 7 The equation  $189n + 356$  models a sandwich shop's costs. The 356 could represent:

- A. Number of sandwiches sold
- B. Fixed daily cost
- C. The cost per sandwich
- D. Number of sandwiches made

- 8 A tap was turned on so that water was flowing out of a tank at a constant rate. The graph shows the volume of water ( $V$ ) left in the tank at the end of  $t$  minutes. The equation of the line is of the form:  $V = mt + c$ , where  $m$  and  $c$  are constants.



- a. Write down the equation of the above line.

$$C = 9000 \quad \text{Rate of water is } 180 \text{ L/min (decreasing)}$$

$$V = -180t + 9000$$

- b. How long did it take to have 1000L in the tank?

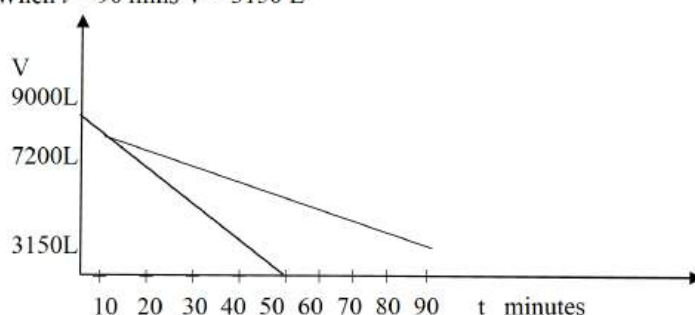
$$1000 = -180t + 9000$$

$$t = 44.444.. \text{ mins}$$

- c. After 10 minutes the tap was turned back so that the water was flowing out at a quarter of the previous rate. On the graph below carefully draw another graph of  $V$  against  $t$  that could represent this new situation. Your new graph should show  $V$  from  $t = 10$  to

$$V = 90 \text{ mins.}$$

After 10 mins volume of water =  $-180(10) + 9000 = 7200 \text{ L}$   
 The rate of flow of the tap =  $\frac{1}{4}$  of  $180 = 45 \text{ L/min}$   
 i.e.  $v = -45t + 7200$   
 When  $t = 90 \text{ mins } V = 3150 \text{ L}$



- 9 A taxi driver charges a standard fare consisting of a \$5 flat fee as well as \$2 per minute while driving. Suppose the driver works for  $t$  minutes. Which equation expresses the amount the taxi driver charges (\$ $C$ ) as a function of time ( $t$  minutes)? 1

- A.  $C = 2 + 5t$  B.  $C = 5 + 2t$   
C.  $t = 2 + 5C$  D.  $t = 5 + 2c$

- 10 The capacity of the fuel tank of Jane's car is 65 litres. She starts driving with a full tank and the car consumes 0.15 litres per km. Which of the following linear equations describes the volume ( $V$ ) in litres of fuel in the tank, after travelling  $k$  km? 1

$$V = -0.15k + 65$$

- 11 a. Make  $y$  the subject of the formula  $3x - 5y + 10 = 0$ . 1

$$3x + 10 = 5y$$

- b. Using your answer in part (a) or otherwise, find the 1

$$y = \frac{3}{5}x + 2$$

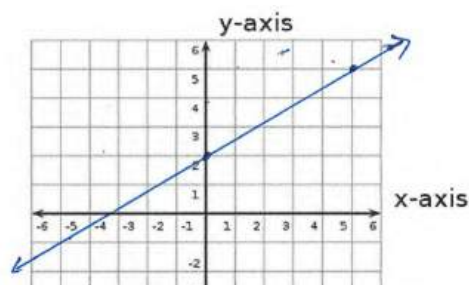
gradient and  $y$ -intercept of the linear equation:

$$3x - 5y + 10 = 0$$

$$\text{gradient (m)} = \frac{3}{5}$$

$$\text{y-int (c)} = 2$$

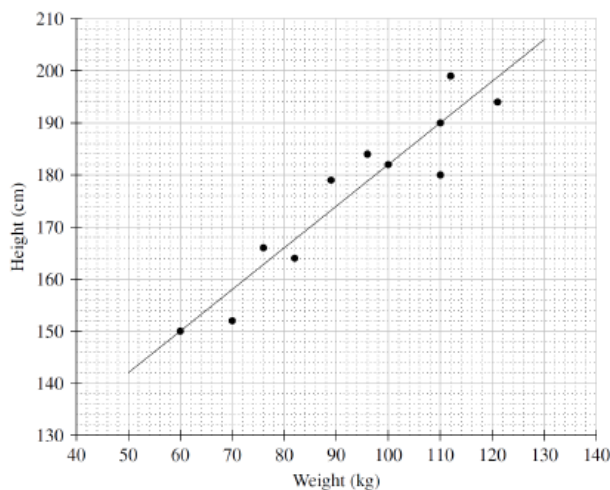
- c. Draw the linear function  $3x - 5y + 10 = 0$ . 2



- 12 The heights,  $H$ , and weights,  $W$ , of several people were taken and a line of best fit drawn. 1

What is the equation of the line of best fit?

- A.  $H = 0.8W + 102$   
B.  $H = 0.8W + 135$   
C.  $W = 0.8H + 102$   
D.  $W = 0.8H + 135$



Straight line passes through (60, 150) and (110, 190)

$$\text{Gradient} = \frac{\text{Change in } H}{\text{Change in } W} = \frac{190 - 150}{110 - 60} = \frac{40}{50} = 0.8$$

$\therefore$  Equation is  $H = 0.8W + c$

Since curve passes through (60, 150),

$$150 = 0.8(60) + c$$

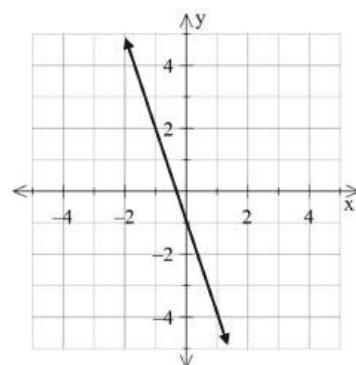
$$150 = 48 + c$$

$$c = 102$$

$\therefore$  Equation is  $H = 0.8W + 102$

- 13 What is the equation of the line shown in the diagram? 1

$$y = -3x - 1$$



14 Make  $y$  the subject in the formula:  $5x - 2y + 4 = 0$ .

$$y = \frac{5}{2}x + 2$$

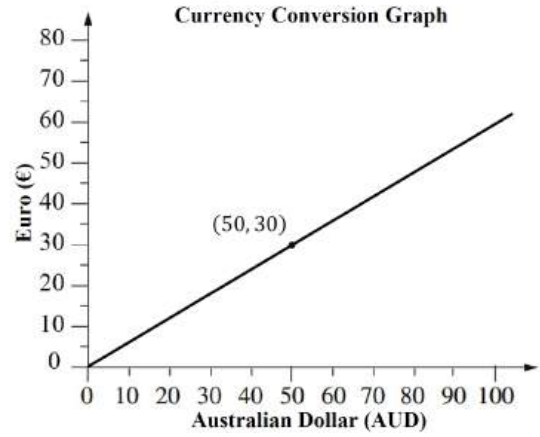
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15 The graph below is used to illustrate the value of currency between the Euro and the Australian Dollar.

Use the graph to convert 340 Australian Dollars to

Euros?

€ 204



1



# Linear Relationships

## Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Which of the following correctly expresses  $y$  as the subject of the formula  $2x - 5y + 3 = 0$ ? 1

$$y = \frac{2}{5}x + \frac{3}{5}$$

- 2 John's car uses 0.12 litres of fuel per kilometre on highway travel. The lowest cost of fuel available is \$1.25 per litre.

- a. What would be the lowest cost for him, in fuel, for a highway trip of 50 km? 1

$$\text{For a trip of 50 km fuel used} = 50 \times 0.12 = 6 \text{ litres}$$

$$\text{Cost} = 6 \times 1.25 = \$7.50$$

- b. Write an equation that gives the lowest cost in dollars ( $C$ ), for fuel, of a highway trip of  $N$  kilometres. 2

$$\text{For a trip of } N \text{ km fuel used} = N \times 0.12 = 0.12N \text{ litres}$$

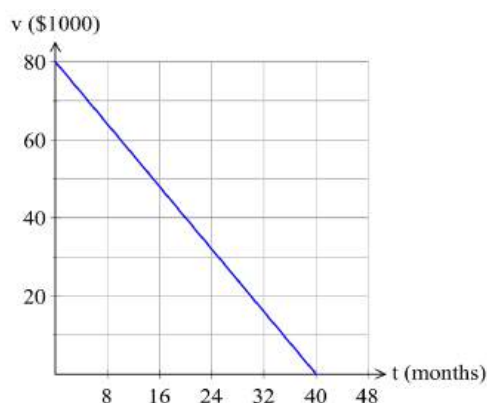
$$\text{Cost} = 0.12N \times 1.25 = 0.15N$$

$$C = 0.15N$$

- 3 A new piece of equipment is purchased by a business for \$80000. Its value is depreciated each month using the graph below. What is the equation of the straight line in terms of  $v$  and  $t$ ?

$$m = -90/40 = -2.25$$

$$v = -2.25t + 80$$



- 4 The weight ( $W$ ) of a metallic ball is directly proportional the cube of its radius ( $r$ ). A ball that has a radius of 2 cm has a weight of 78.96 g. By first finding the value of the constant  $k$ , calculate the radius of a ball that weighs 631.68 g. 3

$$W = kr^3$$

$$78.96 = 8k$$

$$k = \frac{78.96}{8}$$

$$k = 9.87$$

$$W = 9.87r^3$$

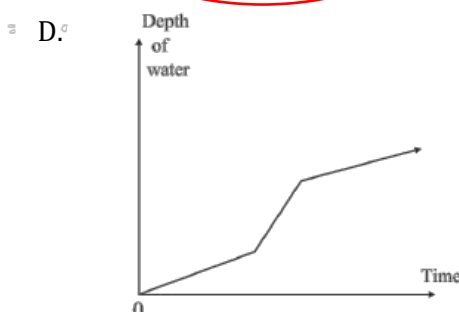
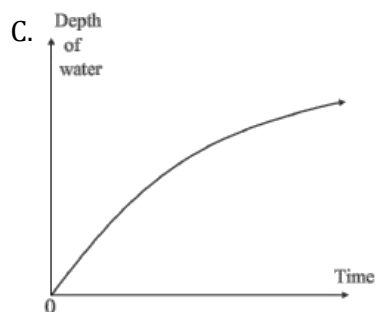
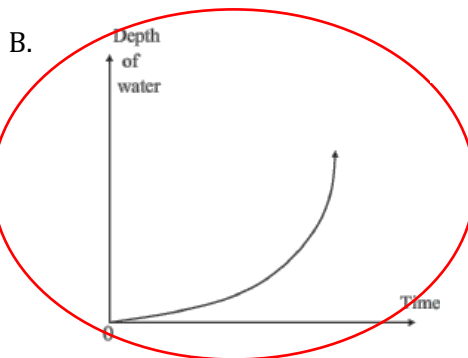
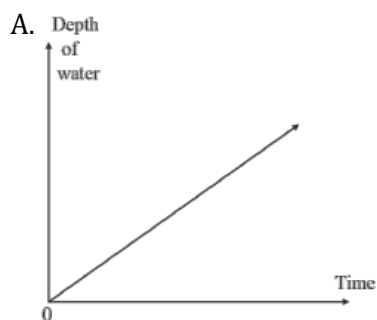
$$\frac{631.68}{9.87} = r^3$$

$$r = \sqrt[3]{64} = 4 \text{ cm}$$

- 5 The value of  $A$  varies directly with  $B$ . It is known that when  $B = 15$ ,  $A = 9$ . What is the value of  $A$  when  $B = 65$ ? 1

39

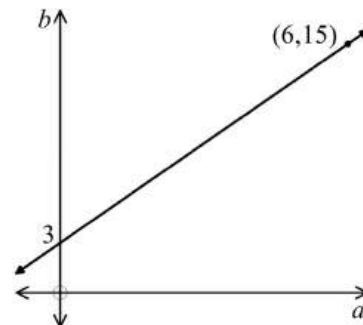
- 6 Water was poured into a container at a constant rate. The container is shown below. Which of the following graphs correctly shows the depth of water in the container as it was being filled? 1



- 7 Find the equation of the line shown. 1

$$m = \frac{12}{6} = 2 \quad y \text{ int} = 3$$

$$\text{Equation: } b = 2a + 3$$



- 8 Jake's watering can is initially filled with 10 litres of water. However, the watering can has a small hole in the base and is leaking at a rate of 0.7 litres per minute. What linear equation describes this situation? 1

$$V = -0.7t + 10$$

A.  $V = -0.7t + 10$

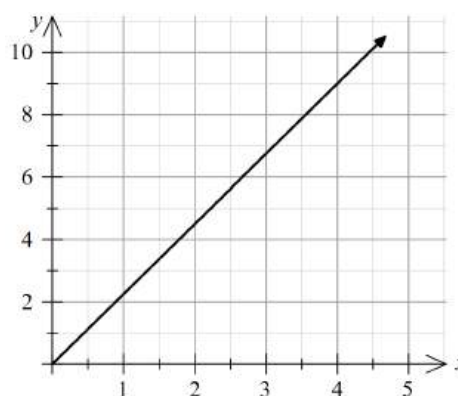
B.  $V = -0.7t - 10$

C.  $V = 0.7t + 10$

D.  $V = 0.7t - 10$

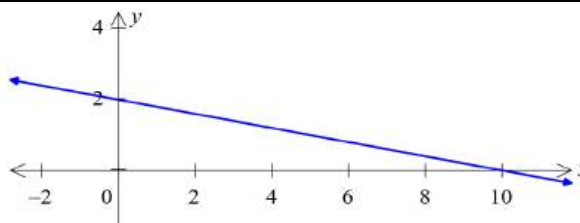
- 9 Find the equation of the line shown. 1

$$y = 2.25x$$



10 What is the equation of the line?

$$y = -0.2x + 2$$



1

11 Which one of the following statements about the line with equation  $15x - 3y = 0$  is NOT true?

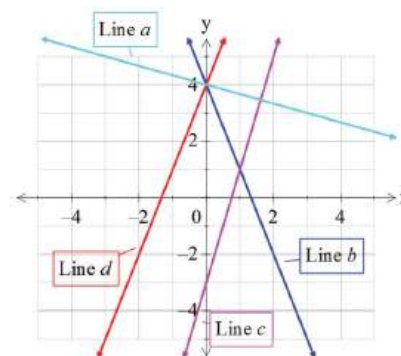
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- A. The line has a gradient of 15.
- B. The line passes through the origin.
- C. The point (1, 5) lies on the line.
- D. The line has a positive gradient.

12 Which line below has the equation  $y = 4 - 3x$ ?

1

- A. Line a
- B. Line b
- C. Line c
- D. Line d



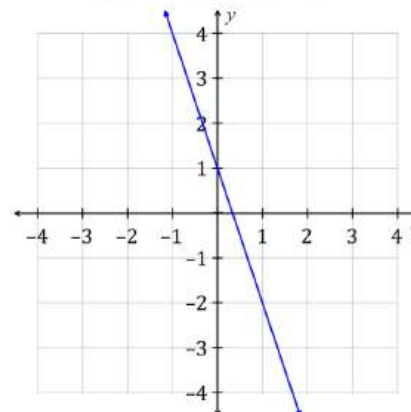
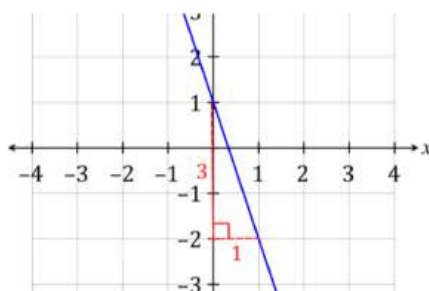
13 Write down the relationship between  $x$  and  $y$ .

1

$$m = \frac{\text{Rise}}{\text{Run}} = -\frac{3}{1} = -3$$

y-intercept is -1

$$y = mx + c = -3x + 1$$



14 Lucas's watering can is initially filled with 4 litres of water. However the watering can has a small hole in the base and is leaking at a rate of 0.2 litres per minute.

a. Write a linear equation in the form  $V = mt + c$  to describe this situation.

1

When  $t = 0$  then  $V = 4$

Leaking at 0.2 litres per minute implies  $m = -0.2$

$$V = mt + c$$

$$V = -0.2t + 4$$

b. What volume of water remains after 150 seconds?

1

Given  $t = 150$  sec or 2.5 min

$$V = -0.2t + 4 = -0.2 \times 2.5 + 4 = 3.5 \text{ L}$$

$\therefore$  Volume remaining after 150 seconds is 3.5 litres.

c. How long would it take for all the water to leak out?

1

Water leaks out when  $V = 0$

$$V = -0.2t + 4$$

$$0 = -0.2t + 4$$

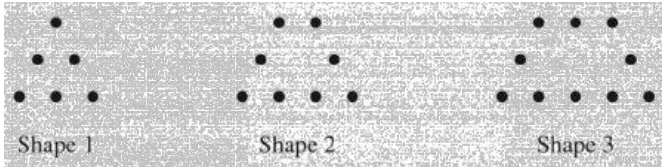
$$0.2t = 4 \quad t = 20 \text{ min}$$

$\therefore$  It takes 20 minutes for the water to leak out.

15

Dots were used to create a pattern. The first three shapes in the pattern are shown.

2



The number of dots used in each shape is recorded in the table.

|                             |   |   |    |
|-----------------------------|---|---|----|
| Shape ( <i>S</i> )          | 1 | 2 | 3  |
| Number of dots ( <i>N</i> ) | 6 | 8 | 10 |

How many dots would be required for Shape 156?

Linear relationship where  $N = 4 + (2 \times S)$   
When  $S = 156$       $N = 4 + (2 \times 156) = 316$

16

Renee went bike riding on a holiday. The hiring charges are listed in the table below:

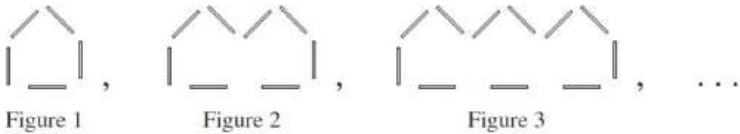
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|                          |    |    |    |    |    |
|--------------------------|----|----|----|----|----|
| Hours hired ( <i>h</i> ) | 1  | 2  | 3  | 4  | 5  |
| cost ( <i>C</i> )        | 18 | 24 | 30 | 36 | 42 |

Which linear equation shows the relationship between *C* and *h*?      $C = 12 + 6h$

17

Sticks were used to create the following pattern.



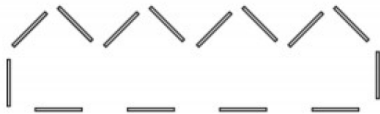
The number of sticks used is recorded in the table.

|                               |   |   |    |
|-------------------------------|---|---|----|
| Figure ( <i>F</i> )           | 1 | 2 | 3  |
| Number of sticks ( <i>N</i> ) | 5 | 8 | 11 |

a.

Draw Figure 4 of this pattern.

1



b.

How many sticks would be required for Figure 100?

1

Since  $N = 2 + 3F$      If  $F = 100$ ,  
 $N = 2 + (3 \times 100) = 302$

c.

Is it possible to create a figure in this pattern using exactly 543 sticks? Show suitable calculations to support your answer.

2

$543 = 2 + 3F$   
 $3F = 541$   
 $F = 180\frac{1}{3}$   
Since *F* is not a whole number, 543 sticks will not create a figure.

18

Suppose  $y = -1 - 2x$ .When the value of *x* increases by 5, what does the value of *y* decreases by?

1

$2 \times 5 = 10$